International Journal of Health Geographics

RESEARCH Open Access

Time-activity and daily mobility patterns during pregnancy and inequalities in air pollution exposure in perinatal outcomes: a cohort study

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Abstract

Background Air pollution exposure assessment during pregnancy often relies on static models based solely on residential addresses. These models may underestimate true exposure and distort assessments of environmental inequalities by neglecting daily mobility. This study aimed to quantify exposure misclassification due to mobility and to assess how it varies across socioeconomic groups in Strasbourg, France.

Methods We analyzed data from 497 pregnant women enrolled in the MOBIFEM cohort. Exposure to nitrogen dioxide (NO_2), particulate matter ≤ 10 µm (PM_{10}), and ≤ 2.5 µm ($PM_{2.5}$), was modeled using six scenarios: one static (residential address only) and five dynamic scenarios incorporating frequented locations and travel itineraries. A local socioeconomic deprivation index was used to stratify the analyses. High-resolution hourly pollutant concentrations were estimated using the ADMS-Urban dispersion model.

Results Dynamic models incorporating mobility data consistently yielded significantly higher mean exposure estimates than the static residential model. The degree of underestimation was greatest among women living in the least deprived neighborhoods (Tertile 1). For instance, the largest discrepancies in NO_2 exposure between static and dynamic models were observed in this group (T-value = -4.72, $p < 0.001$).

Conclusion Relying solely on a residential address leads to substantial underestimation of personal air pollution exposure during pregnancy and may distort the characterization of social inequalities in exposure. In our study area, neglecting mobility disproportionately affected the least deprived women, likely due to longer commuting distances. These findings underscore the importance of incorporating individual mobility data into exposure assessment to improve the accuracy of environmental health risk analyses and inform public health policies.

Keywords Travel pattern, Social inequalities, Air pollution exposure, Pregnant women, Database

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Background

Air pollution exposure is a well-established risk factor for adverse pregnancy outcomes and infant mortality [1–3]. Epidemiological studies have reported potential associations between air pollution levels and adverse birth outcomes, such as low birth weight and preterm birth [4–7].

Among air pollutants, both particulate matter (PM) and nitrogen dioxide (NO_2) have shown consistent associations with birth weight, preterm birth, small-for-gestational-age and impaired postnatal lung function between the ages of 5 weeks and 11 years [6]. Furthermore, numerous studies indicate that air pollution exposure varies with certain socio-economic characteristics such as education, marital status, and/or contextual deprivation [8].

Populations of lower socioeconomic status are disproportionately exposed to air pollutants, primarily due to residential proximity to major emission sources [9–16]. This modifying effect of socioeconomic level can result in both increased exposure and heightened vulnerability among pregnant women, contributing to greater social inequalities in prenatal health.

Despite the growing body of epidemiological research on prenatal exposure to air pollution, most studies are limited by their use of *stationary* exposure models—approaches based solely on the residential address—which may themselves generate or amplify environmental inequalities.

Several modeling techniques have been developed to estimate air pollution exposure in epidemiological studies, including: *(i)* data from the monitoring station nearest the participant's home [17], *(ii)* land-use regression (LUR) [5, 7, 18], *(iii)* dispersion models [19, 20].

Many environmental and epidemiological studies estimate exposure using observed or simulated concentration data at participants' home addresses [21, 22], or at aggregated spatial scales, such as census tracts [23], or indeed at ZIP codes [24]. However, due to limited data, these stationary approaches ignore individual travel pattern (daily mobility), which may result in significant exposure misclassification [25, 26], and potentially bias subsequent statistical analyses [27–35]. Most studies on pregnancy exposure to air pollution have relied on exposure estimates derived from the home address or residential neighborhood (e.g., census tract or ZIP code) [24].

To address this limitation, several approaches have been developed to collect spatiotemporal activity data, including travel pattern interviews and diaries [28, 36], the inclusion of multiple addresses (e.g., home and workplace) or full-day mobility tracking [27, 28] during the temporal window of exposure [21, 31, 37]. More recently, Global Positioning Systems (GPS) [38, 39] and smartphone data [34] have been used to characterize travel patterns. However, such methods are logistically demanding, costly, and burdensome for both researchers and participants, limiting sample sizes.

Assessment of individual exposure is a long and data-intensive process in epidemiological study. Building a database that captures both spatial and temporal variability in pollutant concentrations across all relevant micro-environments would require managing an immense volume of data. Consequently, the daily mobility of pregnant women has rarely been integrated into exposure models, and environmental inequalities may therefore be underestimated [38, 40–42].

One French study demonstrated that ignoring daily mobility can lead to underestimation of prenatal exposure to air pollutants. The study further suggested that this bias (which is inherent to stationary modeling approaches) is more pronounced in socioeconomically deprived neighborhoods, thereby exacerbating environmental health inequalities [43].

Our study addresses two main objectives. First, we examine how, when a dynamic approach is used, exposure misclassification changes as a result of including daily mobility data. Second, we compare the stationary and dynamic approaches in order to investigate the pattern of environmental inequalities trends.

Materiel and methods

Study setting

The study was conducted in the Strasbourg metropolitan region (Eurometropolis) in eastern France, which covers an area of 337.61 km^2 (See Fig. 1). According to the 2016 national census, the population of the Eurometropolis—comprising 33 municipalities—was 491,409 inhabitants. To characterize participants' neighborhoods, we delineated residential areas at the census block level.

This analysis was carried out at infra-municipal level, using small geographical units known as IRIS (*Ilots Regroupés pour l'Information Statistique*). The IRIS is a statistical subdivision used by the French national census in which each residential block is defined based on demographic and geographic criteria. French municipalities are subdivided into one or more such 'islets', each containing approximately 2,000 inhabitants per IRIS. The Strasbourg conurbation has a total of 190 IRIS units.

Cohort characteristic

The MOBIFEM cohort is an observational, multisite study including pregnant women recruited from the two University Hospitals of Strasbourg. Data on pregnancy outcome and individual characteristics were obtained from pregnancy registers, clinical records and electronic questionnaires. Recruitment occurred between February 2021 and December 2021, and targeted women from their first trimester onward, following their prenatal ultrasound appointments. Participants completed

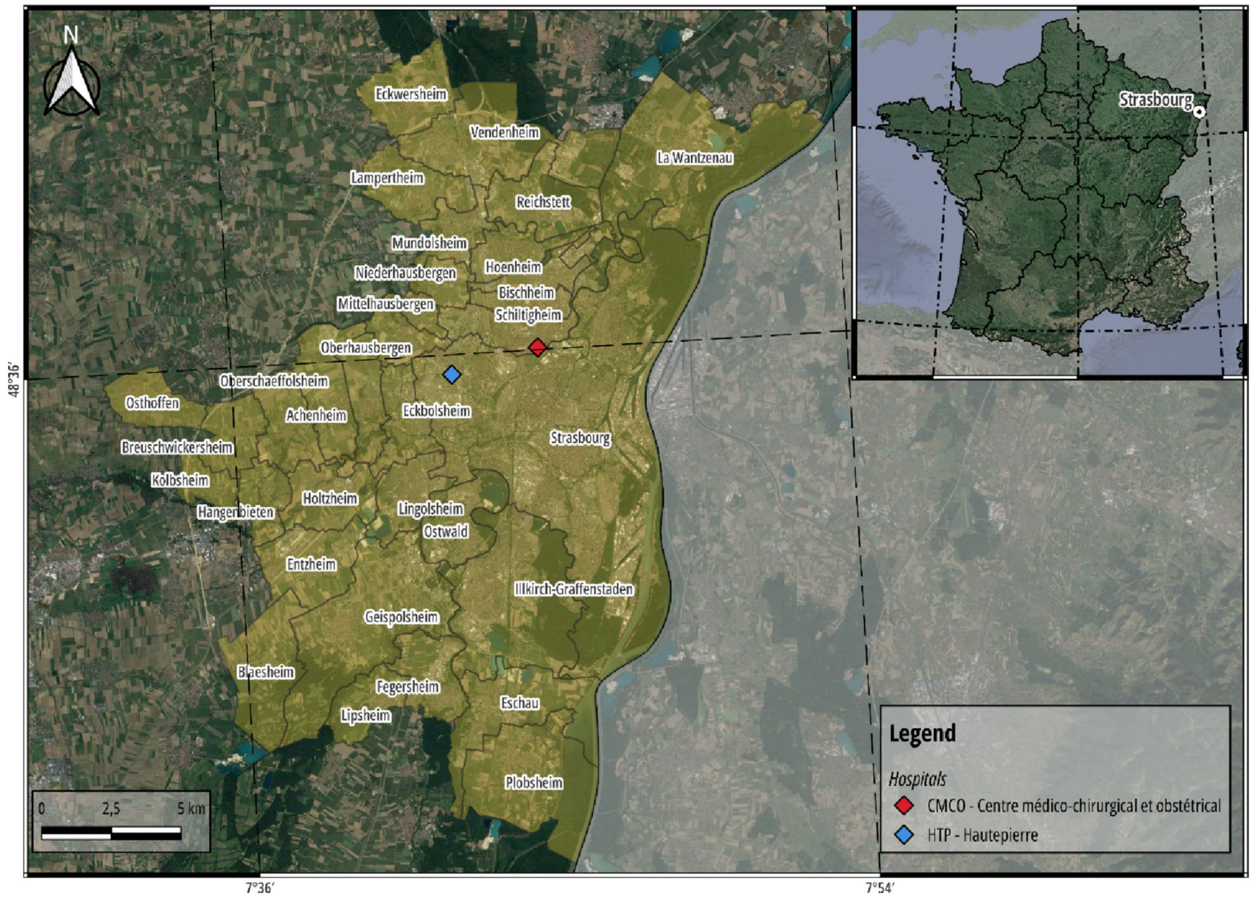

Fig. 1 Location of the Eurometropole of Strasbourg in France and the two university hospitals where the pregnant women in the MOBIFEM cohort were recruited

three online questionnaires at different stages of pregnancy, covering socio-demographic characteristics, travel behavior and lifestyle factors. A detailed description of the recruitment protocol has been published elsewhere [44]. For each participant, the pregnancy period was reconstructed on a daily, weekly, and trimester basis using the date of birth and gestational age. Both hourly air pollutant concentrations at each itinerary point, as well as residential and occupational locations were averaged by trimester, week, and day of pregnancy. Trimesters were defined as follows: First trimester: weeks 1–13; Second trimester weeks 14–26; Third trimester week 27 until birth.

Maternal sociodemographic characteristics

Individual data, including demographic and socioeconomic characteristics (e.g., maternal age, infant sex, occupation, education, and employment status), were collected using a standardized data form. Full details are available in a previous publication [44]. In this study, sex is considered as a biological characteristic, typically assigned at birth based on observable physical traits such as external genitalia. This binary classification (male/female) reflects physiological and chromosomal differences, though it may not capture all variations, including 'intersex' traits or differences in sex development.

Socioeconomic deprivation index

Neighborhood-level socioeconomic context was characterized using the 'EMS index,' a local deprivation index specifically developed for the Eurometropole of Strasbourg ([45]– [46]). This index was constructed from 2012 national census data (INSEE) through successive Principal Component Analyses (PCA) to identify 15 key variables related to housing, family structure, employment, and income. For the purposes of this study, the deprivation index was categorized into tertiles—least deprived, moderately deprived and most deprived—to compare exposure levels across socioeconomic groups.

Pregnancy exposure assessment

Air pollutant concentration data

Hourly ambient concentrations of three pollutants—nitrogen dioxide (NO_2), particulate matter ≤ 10 µm (PM_{10}), and particulate matter ≤ 2.5 µm ($PM_{2.5}$)—were provided by ATMO Grand-Est, the official regional air quality monitoring agency accredited by the French Ministry for the Ecological Transition.

These concentrations were estimated using the ADMS-Urban dispersion model (Cambridge Environmental Research Consultants), a state-of-the-art system designed for high-resolution (meter-scale) air quality modeling in complex urban environments. The model's reliability is attested by its widespread use in regulatory and research applications, including the health impact and exposure studies for which it was designed, and its adoption by the French Approved Air Quality Monitoring Associations (AASQA) [47].

The model's methodology integrates a comprehensive inventory of urban emission sources (traffic, industrial, and residential) and explicitly accounts for urban morphology through advanced modules for features like street canyons. This is coupled with hourly meteorological data and background pollution levels derived from measurements at the regional monitoring station network. To ensure the accuracy of the final concentration maps, the model is routinely calibrated against these monitoring station measurements using the FAIRMODE European methodology.

Daily mobility assessment

Participant mobility was assessed through a structured travel behavior questionnaire administered at three stages of pregnancy: enrollment, between weeks 22 and 28, and between weeks 32 and 38. Participants report their typical travel patterns over the previous three months, including both working and non-working days.

Data collected included residential and work addresses, and up to five frequently visited locations, three on weekdays and two on weekends (e.g., leisure sites, supermarkets, schools).

Details were recorded for each trip, such as departure and arrival times, point of departure, duration of stay at destination, transport mode, and estimated travel time. We built the travel patterns during pregnancy as follows:

Phase 1: travel time matrix computation

Based on the reported data, individual daily mobility to frequented locations was used to compute multiple realistic travel itineraries for each trip. Assuming individuals select the shortest or fastest routes, we generated between two and four alternative itineraries per origin–destination pair using the Dijkstra algorithm, implemented via the st_network_path() function in the sfetworks R package.

Six models were used to estimate exposure during the first two trimesters of pregnancy.

- Mod. 1 assumes pregnant women remain at home all day.
- Mod. 2 considers the three most-frequently-visited locations at specific time slots reported by the participant.
- Mod. 3 incorporates the shortest travel paths (using the Dijkstra algorithm with time-weighted in minutes) between frequented locations; where time slots were missing, residence was assumed.
- Mod. 4 is similar to Mod. 3, but based on shortest distances (Dijkstra algorithm with distance-weighted in meters).
- Mod. 5 adjusts time-weighted itineraries for heavy traffic conditions (time weights increased by 30% to 60%, depending on the road and on detected use of the 'car' mode of transport).
- Mod. 6 prioritizes low traffic routes favoring pedestrian and cycling paths (if relevant modes were reported) and applies distance-weighting on certain axes.

Phase 2: identification and extraction of representative point to modelled air pollution concentration

Beyond all locations frequented (workplace, residence addresses, leisure location.), the objective of this step was to identify a set of points along each trip, based on the spatial and temporal metrics of each participant for each scenario. Each route segment was divided into points spaced 50 m apart, positioned along the central lane for motorized vehicles and public transport. For cycling routes, a 3 m buffer was applied from the lane center; for walking routes, a 5 m buffer was used. This process yielded a dense spatial network of points covering all frequented locations and travel itineraries under each modeling scenario, allowing subsequent estimation of pollutant concentrations.

Phase 3: modeling hourly exposure to air pollutants

Hourly pollutant concentrations at each geolocated point (residential, occupational, and travel) were extracted from the ADMS-Urban model. Using spatiotemporal mobility data, we estimated 24-hour concentration profiles for each participant across all pregnancy periods.

Phase 4: reconstruction of daily exposure

This phase combined participants' time–activity patterns with hourly pollutant concentrations to compute individualized dynamic exposure profiles.

Two modeling approaches were compared:

- Stationary model (Model 1): Assumes women remained at home all day.
- Dynamic models (Models 2–6): Weighted pollutant concentrations according to time spent at each location and mode of transport.

Daily exposure estimates were aggregated to weekly and trimester averages, producing exposure profiles at daily, weekly, monthly, trimester, and full-pregnancy resolutions.

Database construction

Given the volume of data (billions of concentration values), we chose to develop a SQLite relational database to efficiently process and link pollutant concentrations with individual mobility trajectories. Advantages of this approach included:

- Improved data access using host and query languages.
- Reduced data redundancy.
- Simplified replication.

The database stored hourly pollutant concentrations across the entire Eurometropolis. Multiple indexes were created to optimize SQL query performance. Data extraction and linkage to participants' mobility data were performed in Python using pandas to reconstruct individual exposure time series for each pregnancy hour.

Statistical analysis

Descriptive statistics (mean, standard deviation, box plots) were first computed to summarize exposure distributions across the six modeling scenarios (Models 1–6). Independent *t*-tests were conducted to assess differences in mean exposure estimates between the different methods of characterization of exposure. The significance level was set at $p < 0.05$. Paired T-tests were also performed to compare within-subject exposure differences across modeling approaches.

Lastly, Bland-Altman plots were used to measure agreement between texposure estimation methods. This analysis helped assess interchangeability, and evaluate how much they deviated from one another.

Statistical analyses as descriptive statistics and paired-difference tests were performed using R, and the map representations with Qgis software.

Result.

Study population description

A total of 497 participants were recruited during the first trimester prenatal ultrasound appointments at two sites: Hautepierre Hospitals (63%) and the CMCO (37%). The mean maternal age of the cohort was 31 years. Main descriptive characteristics are summarized in Table 1. Among the participants, 38% of the expectant mothers were nulliparous, 41% primiparous, and 22% multiparous. Regarding individual socio-economic status, most women were employed (39%), inactive (28%), or occupied intermediate professions (23%).

Approximately 70% of participants lived in apartments, while the rest resided in detached houses. Average gestational age at birth was 38.8 weeks of amenorrhea (SD: 4.1 weeks). Sixteen infants were born prematurely before 36 weeks of amenorrhea, one before 32 weeks, and three before 28 weeks. Mean birth weight was 3,218 g (SD: 606 g). 24 infants had low birth weight (< 2,500 g), 3 had very low birth weight (< 1,500 g), and 4 had extremely low birth weight (< 1,000 g).

Environmental characteristics of residential areas

The mean residential NO2 concentration across participants was 15.45 µg/m3, ranging from 7.26 µg/m3 in Griesheim to 20.72 µg/m^3 in Schiltigheim (Strasbourg average: 17.43 µg/m3). Mean PM_{10} concentration was 15.88 µg/m^3, ranging from 13.01 µg/m^3 in Saessolsheim to 17.27 µg/m^3 in Schiltigheim (Strasbourg average: 16.46 µg/m^3). Mean $PM_{2.5}$ concentration was 9.12 µg/m^3, varying from 6.50 µg/m^3 in Innenheim to 10.82 µg/m^3 in Fulgriesheim (Strasbourg average: 9.48 µg/m^3).

Table 1 Main descriptive characteristics of pregnant women included in the cohort study

Characteristics	Study group (497)
Age (years (mean +/- SD))	31yo +/- 5,3
Location of inclusion (n(%))	
Hautepierre	312 (62%)
CMCO	185 (37%)
Parity (n (%))	
Nulliparous	187 (38%)
Primiparous	202 (41%)
Multiparous	108 (22%)
Socio-professional categories (n(%))	
Small employers and own account workers	8 (2%)
Higher managerial, administrative and professional occupations	35 (7%)
Employee	182 (39%)
Inactive	131 (28%)
Workers	4 (1%)
Intermediate occupations	110 (23%)
Type of housing (%)	
Appartment	70%
House	30%
Gestational age issue (mean +/- SD)	38,8sa +/- 4,1
Preterm birth (< 36sa) (n(%))	16 (3%)
Very preterm birth (< 32sa) (n(%))	1 (0,2%)
Extremely preterm birth (< 28sa) (n(%))	3 (1%)
Birth weight (g) (mean +-SD)	3218 g +/−606
Low birth weight (< 2500 g) (n(%))	24 (5%)
Very low birth weight (< 1500 g) (n(%))	3 (1%)
Extremely low birth weight (< 1000 g) (n(%))	4 (1%)

While 398 participants (80%) completed the first-trimester questionnaire and provided their residential information, only 150 described their most-frequented locations and itineraries. The response rate declined further, with only 50 participants providing mobility data during the second and third trimesters. Given the reduced response rate in later trimesters, we assumed participants resided at their last reported first-trimester address for the remainder of pregnancy. Consequently, the static analyses res based on residential address were conducted for all three trimesters, while dynamic analyses involving travel itineraries were restricted to the first trimester.

Patterns of exposure to NO_2, PM_{10} and $PM_{2.5}$ during pregnancy, across the 6 scenarios

Table 2 presents mean concentrations of NO_2, PM_{10} and $PM_{2.5}$ across trimesters (at residential addresses (Mod.1) only) and during the pregnancy period (all scenarios). Regardless of the pollutant, average concentrations were relatively consistent across all 6 scenarios.

Figure 2 summarizes air pollutant distributions for each modeling approach (Mod.1 to Mod.6). When frequented locations (Mod.2) and travel-related exposures (Mod.3–Mod.6) were considered, exposure estimates tended to increase slightly, indicating modest overexposure.

Comparison between 'Static' and 'Dynamic' exposure estimates

Paired difference analyses to compare exposure estimates between static residential model (Mod.1) and the five dynamic models (Mod.2- Mod.6)

Table 3 shows statistically significant results. More precisely, results of the paired difference tests tend to show that the level of exposure to the different pollutants (NO_2, PM_{10} and $PM_{2.5}$) is underestimated when only residential address is considered in characterizing pregnancy exposure, ignoring frequented locations (Mod.2), daily mobility (Mod.4 and Mod.6) and travel mode (Mod.3 and Mod.5).

For NO_2 exposure, highly significant differences were observed between Mod.1 and all other scenarios, with t-values ranging from −4.05 to −7.25. For PM_{10} exposure, the differences were also highly significant, with t-values ranging from −2.69 to −6.87. And for $PM_{2.5}$ exposure, no significant difference was observed between the exposure estimates of Mod.1 and those of Mod.2 ($p > 0.05$, t = −1.27). However, highly significant differences were found when comparing Mod.1 to the other scenarios (Mod.3 to Mod.6), with t-values ranging from −2.04 to −5.84.

Comparison among dynamic models: daily mobility (Mod.4, Mod.6) and travel mode (Mod.3, Mod.5)

Table 4 describes variations in exposure estimates according to different daily mobility scenarios (Mod.4, Mod.6, Mod.3, Mod.5). All paired differences of exposure estimates according to different scenarios were significant, demonstrating that incorporating mobility characteristics affects exposure estimates. Across all pollutants (PM_{10}, $PM_{2.5}$, $NO_{2})$, exposure underestimation was greatest for Mod.6, followed by Mod.5 and Mod.3. Mod. 4 consistently produced the highest estimated concentrations. Across all pollutants, differences between Mod. 5 and Mod. 6 were minimal.

Social inequalities in exposure during pregnancy

Figure 3 shows differences in NO_2, PM_{10} and $PM_{2.5}$ concentrations according to tertiles of socioeconomic deprivation. Across all three pollutants, inequalities in residential exposure (Mod.1) were evident. Women residing in moderately deprived census blocks (Tertile 2) experienced the highest mean exposures, compared with those in both the least and most deprived areas (Tertiles 1 and 3).

Table 2 Concentrations of NO_2, PM_{10} and $PM_{2.5}$ (Average and standard deviation, expressed in µg/m^3) estimated for 6 exposure scenarios, across pregnancy trimesters

	NO_2		PM_{10}		$PM_{2.5}$	
	Mean	**SD**	**Mean**	**SD**	**Mean**	**SD**
Static approach						
T1	15.98		16.58		9.63	
T2	14.33		15.38		8.55	
T3	16.27		16.25		9.83	
Mod.1	15.46	4.18	16.06	2.71	9.29	2.66
Dynamic approaches						
Mod.2	16.52	3.57	16.71	2.66	9.66	2.58
Mod.3	16.80	5.75	17.01	2.71	9.88	2.62
Mod.4	17.04	6.34	17.04	2.82	9.82	2.65
Mod.5	17.12	6.06	17.14	2.75	9.92	2.67
Mod.6	15.87	5.25	16.82	2.75	9.77	2.61

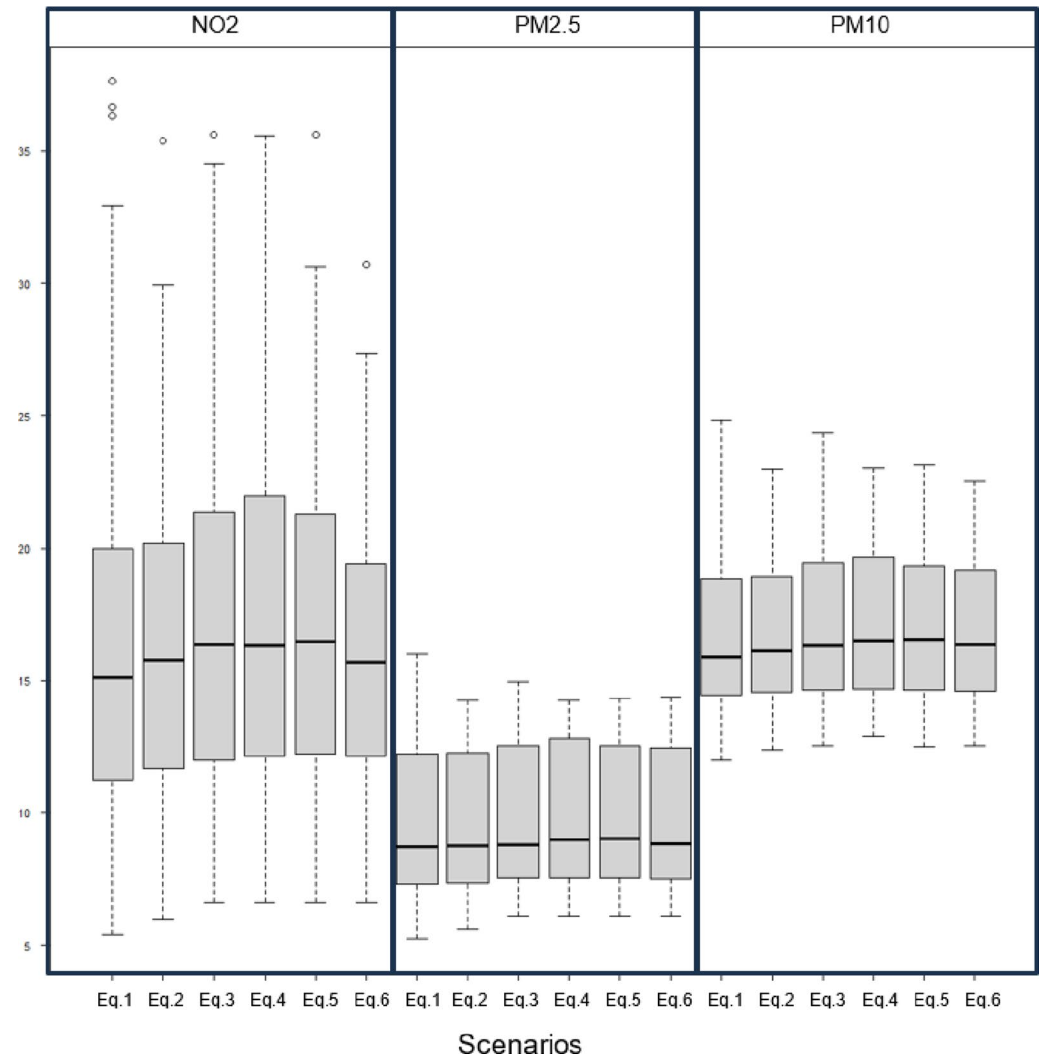

Fig. 2 Distribution of concentrations of NO_2, PM_{10}, $PM_{2.5}$ estimated using the stationary approach (Mod.1, residential address only) and dynamic models (Mod.2-Mod.6)

Table 3 Paired differences in NO_2, PM_{10} and $PM_{2.5}$ exposure (µg/m^3) estimated between different scenarios

Mod.1 compared to :	Mod.2		Mod.3		Mod.4		Mod.5		Mod.6	
	T value	*P* value	T value	*P* value	T value	*P* value	T value	*P* value	T value	*P* value
PM_{10}	−2.69	< 0.01	−6.31	< 10 − 05	−6.87	< 10 − 05	−3.09	< 0.01	−4.79	< 10 − 05
$PM_{2.5}$	−1.27	NS	−5.84	< 10 − 05	−5.56	< 10 − 05	−2.04	< 0.05	−3.52	< 0.001
NO_2	− 4.05	< 10 − 05	−6.84	< 10 − 05	−7.25	< 10 − 05	−4.04	< 0.001	−5.71	< 10 − 05

A similar pattern was observed when frequented locations were included in exposure estimation (Mod.2; see Fig. 3b).

The difference between the estimated exposure from the statistic residential model (Mod.1) and from dynamic models (Mod.2-Mod.6) varied by socioeconomic tertile.

An analysis of the paired differences between the static model (Mod. 1) and the dynamic models (Mod. 2, 3, and 4) reveals that the magnitude of exposure misclassification varies significantly across socioeconomic groups, as shown in Table 5. The results consistently indicate that the underestimation of exposure is most pronounced for pregnant women living in the least deprived neighborhoods (Tertile 1).

When comparing the static Mod. 1 to the dynamic Mod. 2, a statistically significant difference in NO_2 exposure was observed only in Tertile 1 (t-value = −3.66, $p < 0.001$). The differences for Tertile 2 and Tertile 3 were not significant.

The largest difference for NO_2 exposure occurred when comparing Mod. 1 to Mod. 3, again within Tertile 1, with a (t-value of −4.72, $p = 10^{-5}$).

Table 4 Paired differences in NO_2, PM_{10} and $PM_{2.5}$ exposure estimated (in µg/m^3) among dynamic models (Mod.3- Mod.6)

	Mod.3		Mod.4		Mod.5	
	T value	***P* value**	**T value**	***P* value**	**T value**	***P* value**
	NO_2					
Mod.4	−3.16	< 0.01		–		–
Mod.5	−4.36	< 10^{-05}	−5.96	< 10^{-05}		
Mod.6	−3.22	< 0.01	−3.96	< 0.001	−2.39	< 0.01
	PM_{10}					
Mod. 4	−3.30	< 0.001		–		–
Mod.5	−4.14	< 10^{-05}	−6.04	< 10^{-05}		
Mod.6	−3.07	< 0.01	−4.21	< 10^{-05}	−1.98	< 0.05
	$PM_{2.5}$					
Mod.4	−3.10	< 0.01				
Mod.5	−4.39	< 10^{-05}	−5.89	< 10^{-05}		
Mod.6	−2.96	< 0.01	−4.03	< 0.001	−2.00	< 0.05

For PM_{10} and $PM_{2.5}$, the dynamic models also yielded significantly higher exposure estimates than the static model across all tertiles, with the largest discrepancies consistently in Tertile 1 (e.g., t-value = −4.25 for PM_{10}; t-value = −3.6256 for $PM_{2.5}$ for Mod. 1 vs. Mod. 3). Negative t-values indicate systematic underestimation by the static model.

These findings suggest that neglecting daily mobility results in the greatest exposure misclassification among women from the least deprived areas. This pattern likely reflects different lifestyle and residential differences; women in less deprived census blocks often live in suburban areas further from the urban core, resulting in longer and more frequent commutes by car. This increased travel time through potentially more polluted central areas could explain why their exposure is more significantly underestimated when only their residential address is considered.

Bland–Altman plots illustrating the relationship between Mod.1 and the dynamic models (Mod.3, Mod.4) are provided in the Supplementary Materials (Appendix). The coefficient reflects the extent to which exposure estimates differ between two models. The coefficients quantify the magnitude of exposure misclassification when daily mobility (Mod.2, Mod.3) and travel mode (Mod.4) are not incorporated. A coefficient close to 0 indicates a lower degree of misclassification in NO_2 exposure estimates.

Discussion

Our study confirms the presence of environmental inequalities in the Strasbourg metropolitan area, regardless of whether maternal mobility during pregnancy was accounted for. We observed a positive relationship between socioeconomic deprivation and ambient air pollution: on average, NO_2, PM_{10} and $PM_{2.5}$ concentrations are lower in the least deprived tertile and increased progressively across the moderate and most deprived tertiles. This pattern aligns with previous IRIS-level research conducted in the Strasbourg metropolitan area, using an alternative deprivation index, which identifies an inverse-U curve linking air pollution exposure to neighborhood deprivation [48, 49]. Although average pollutant concentrations have drastically fallen due to land use planning and environmental regulations, persistent inequalities remain and exhibit similar spatial patterns. These disparities likely reflect the city's historical urban development, economic and political structures, demographics, degree of industrialization, and urban morphology. For instance, urban structure tends to favor a higher concentration of affluent populations near the city center, whereas lower socioeconomic groups are more likely to reside in suburban and peri-urban areas [50].

Unlike findings from many American studies, European research suggests that the nature and direction of environmental inequalities are context-dependent, influenced by local spatial and socioeconomic dynamics. Some studies report higher pollutant exposures among low socioeconomic status (SES) populations [51, 52], others among middle-SES groups [48], while some even document an inverse association [53].

Recent French studies have confirmed that the relationship between air pollutant concentrations and socioeconomic characteristics is highly context-dependent. The associations observed at census block level vary across study areas; for instance, patterns in the Lille metropolitan area differ from those in Paris while in Lyon, no clear socioeconomic gradient is apparent. In that city, census blocks with intermediate levels of deprivation show the highest exposure to traffic-related air pollution, which is consistent with findings from the Strasbourg metropolitan area [48, 54, 55].

However, most of these studies have assessed socioeconomic deprivation and air pollution exposure based solely residential addresses – disregarding population mobility. Given the profound transformations in daily life

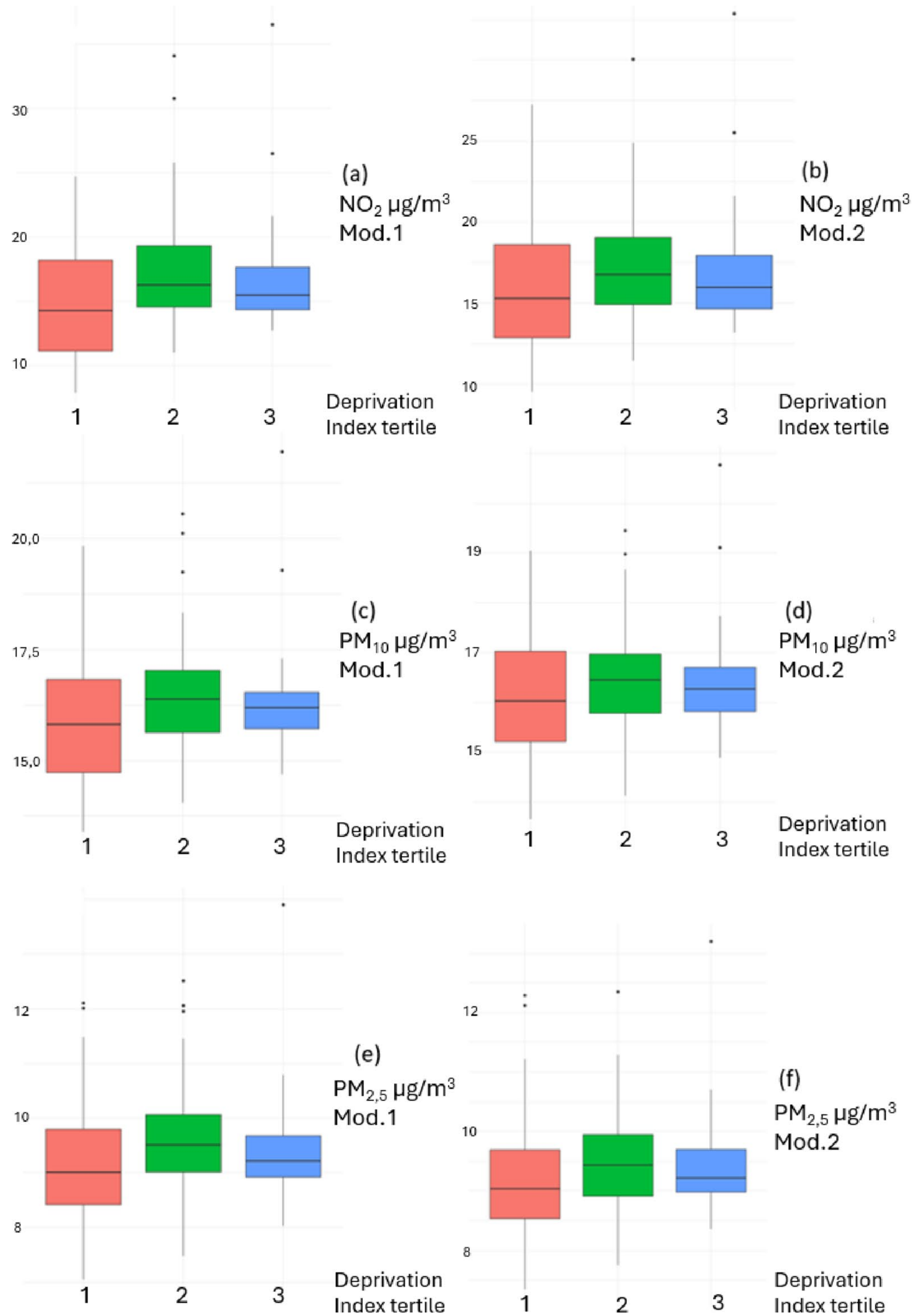

Fig. 3 Distribution of NO_2 concentrations (μg/m^3) estimated from Mod.1 (**a**) and from Mod.2 (**b**), at the residential addresses, by tertile of socioeconomic deprivation during the pregnancy; Distribution of PM_{10} concentrations (μg/m^3) estimated from Mod.1 (**c**) and from Mod.2 (d), at the residential addresses, by tertile of socioeconomic of deprivation during the pregnancy; Distribution of $PM_{2.5}$ concentrations (μg/m^3) estimated from Mod.1 (e) and estimated from Mod.2 (f), at the residential addresses, by tertile of socioeconomic of deprivation during the pregnancy

Table 5 Paired differences in PM_{10} exposure estimated (µg/m^3) during first trimester comparing Eq. 1. With different scenarios (Mod.2, Mod.3, Mod.4,) by tertile of socioeconomic deprivation

	Tertile 1 (less deprived)		**Tertile 2**		**Tertile 3 (most deprived)**	
	T value	***P* value**	**T value**	***P* value**	**T value**	***P* value**
	Mod.1 versus Mod.2					
NO_2	−3.66	< 0.001	− 0.20	NS	−1.34	NS
PM_{10}	−1.86	NS	− 0.49	NS	−1.32	NS
$PM_{2.5}$	−0.78	NS	− 0.91	NS	−0.80	NS
	Mod.1 versus Mod.4					
NO_2	−4.03	< 0.001	−3.24	< 0.01	−2.42	< 0.05
PM_{10}	−2.60	< 0.01	−3.89	< 0.001	−2.38	< 0.05
$PM_{2.5}$	−1.61	NS	−3.19	< 0.01	−2.06	< 0.05
	Mod.1 versus Mod.3					
NO_2	−4.72	10−05	−2.84	< 0.01	−2.86	< 0.01
PM_{10}	−4.25	< 0.0001	−3.40	< 0.001	−2.58	< 0.01
$PM_{2.5}$	−3.6256	< 0.001	−3.38	< 0.001	−2.67	< 0.01

that have increased travel for work, leisure, and errands, it is reasonable to question the impact of mobility on exposure estimates. This is the additional facet explored by this project, and indications are that to ignore population mobility (regardless of scenario) will lead to underestimation of the level of exposure to air pollutants. This study addressed that issue directly, showing that ignoring mobility—regardless of the scenario applied—leads to a systematic underestimation of exposure levels. Our findings further reveal that this bias is greatest among pregnant women residing in the least deprived census blocks, likely reflecting their longer commuting distances compared to those living in more deprived areas. A previous study conducted in Paris reported the opposite pattern: taking mobility into account increased the NO_2 exposure levels of pregnant women living in the most deprived areas [50]. These contrasting results highlight the influence of spatial organization in the metropolitan area.

Both the study design and the data used in this present work present notable strengths and limitations.

Our method was designed to study the characteristics of local environment without having to rely on specialized exposure-measurement devices. By adapting to the spatial and socioeconomic features of the study area, it allows detailed analysis of mobility-related exposure dynamics. This flexible and reproducible approach can be applied in other territories to facilitate comparative research and to guide targeted interventions. Future work can usefully explore patterns in other contexts, other cities, in other countries, to develop the underlying framework and examine changes over time. As urbanization and economic development progress alongside improvements in air quality, the current configuration of environmental inequality may change over time. In recent years, with public policy reform building pressure to decrease the use of thermic motors, fossil fuels, etc., public awareness of air pollution has increased. These shifts could further alter the relationship between SES and air pollution exposure. Environmental justice theory in the United States [56, 57] and elsewhere [58, 59] highlights the concept of sacrifice zones – areas lacking political power and exposed to disproportionate environmental risk [60]. This is a critical aspect of how relationships might shift over time and will depend in part on inequities in political power.

Our study focused exclusively on external ambient concentrations, as is common in environmental epidemiology, and did not consider near-source exposures such as environmental tobacco smoke, indoor air pollution, or emissions from solid fuel use for cooking or heating) [61–73]. Setton et al., (2008) and Marshall et al. (2006), demonstrated that while time spent at home accounts for most exposure differences across census tracts, time spent at workplaces contributes substantially to within-tract variability [65, 74]. Additionally, outdoor and indoor air pollution levels are correlated, with an estimated infiltration factor of 0.66 for NO_2 found in various indoor environments in Sweden [75]. Because individuals in more deprived populations often live in less well-insulated dwellings, indoor exposure may account for a larger share of their total exposure. We can hypothesize that considering indoor air pollution would exacerbate environmental inequalities. This is especially true of the most deprived areas, and future work could provide important additional insights regarding these factors [76–78].

The moderate sample size of our cohort may have limited the statistical power to detect certain inequalities. Furthermore, our dynamic exposure models, which account for mobility, were restricted to the first trimester due to low response rates for subsequent questionnaires. This represents a significant limitation and a potential source of bias. Travel patterns are likely to change throughout pregnancy, particularly with the reduction or cessation of work-related commutes in the third trimester. By relying on mobility data from what is arguably the most active period of pregnancy, our dynamic models may overestimate the average exposure misclassification when compared to the static residential model across the entire gestational period. While the first trimester is a critical window of vulnerability for fetal development, future longitudinal studies with more complete mobility data are needed to confirm these findings and understand how these patterns evolve across all three trimesters. On the other hand, our models did not consider multi-pollutant effects. Furthermore, our analyses did not consider multi-pollutant interactions, and unmeasured environmental factors correlated with NO_2 or PM levels could have influenced the results. Future studies involving larger cohorts should incorporate multi-pollutant models to capture possible synergistic interactions between different ambient pollution.

In France, as in other countries, measures and interventions aimed at improving air quality are regularly implemented. While their aggregate environmental and health benefits are often quantified, few studies have evaluated how these benefits are distributed across socioeconomic groups. Available evidence suggests that public health measures may inadvertently reinforce social gradients, as more advantaged groups tend to adopt preventive behaviors earlier and benefit sooner from health-promoting interventions. Without policies explicitly targeting disadvantaged populations, such initiatives risk widening social and territorial health inequalities. Benmarhnia et al. illustrated this dynamic in their evaluation of London's Low Emission Zone (LEZ), showing that the intervention altered the spatial distribution of air pollution to the detriment of the most vulnerable populations [79]. Similarly, findings from the Equit'Area project in Lille revealed that decreases in NO_2 concentrations were smaller in the most deprived IRIS compared to the most affluent ones [55]. These observations underscore the need to integrate equity considerations into air quality management. The promotion of healthy urban environments should adopt a *proportionate universalism* approach–according to which health initiatives for the entire population should be proportional to pre-existing social disadvantage. In this context, the present work could be viewed as an essential step in thinking about designing strategies aimed at improving air quality.

In conclusion, this study finds that for three major air pollutants (NO_2, PM_{10} and $PM_{2.5}$), average ambient exposure levels in the Eurometropole of Strasbourg are higher for higher-SES populations, and also higher when the pregnant woman's mobility and time-space of activity are considered; this implies that exposure will not be captured by static residential models. These results, consistent across multiple mobility scenarios and based on high-resolution exposure modeling, confirm that residential-only approaches substantially underestimate true exposure. The disparities for NO_2 and PM_{10} tend to be larger than for $PM_{2.5}$. Although this direction of inequality contrasts with patterns reported in much of the environmental justice literature, it remains consistent with European findings emphasizing context-specificity. Overall, these insights carry important implications for local public policy and may inform future comparative research across metropolitan areas with differing socioeconomic and spatial configurations.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12942-025-00436-4.

Supplementary Material 1

Acknowledgements
Valentin Simoncic is recipient of a doctoral grant from the Fondation de France and the Grand Est Region.

Author contributions
VC: Conceptualization, methodology, software, formal analysis, investigation, writing—original draft, visualization. RW: methodology, software, validation, writing—review and editing, visualization. PD: validation, resources, visualization. NS: validation, resources, visualization. CS: validation, resources, visualization. LH: validation, investigation, visualization. SD: Conceptualization, methodology, validation, formal analysis, writing—review and editing, visualization, supervision. WKT: Conceptualization, methodology, validation, formal analysis, writing—original draft, visualization, supervision, project administration, funding acquisition. All authors read and agreed to the published version of the manuscript.

Funding
First author is recipient of a doctoral grant from the Fondation de France and the Grand Est Region.

Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval
Our research was approved by the Commission de Protection des Personnes (CPP) Ile de France XI (Paris) on 28 december 2020 (File reference No. 20 080–42137 IDRCB 2020- A02581- 38). The Agence Nationale de Sécurité du Médicament was informed of it on 15 December 2020. The trial registration numbers are NCT04705272, NCT04725734.

Human ethics and consent to participate declarations
Participants were informed of the research objectives, procedures, potential risks and benefits. Their anonymity and the confidentiality of their personal information were preserved, in strict compliance with all subjects' right to privacy. Formed consent was obtained from all participants included in the study.

Competing interests
The authors declare no competing interests.

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Received: 25 June 2025 / Accepted: 11 November 2025
Published online: 13 December 2025

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